

Assembly at the Kernel

One Joint Law, One Groupoid, One Symmetry Group:
How the Kernel Forces Ignorance, Observables, Duration, Cone and Physical
Distance
on a Physical Slice of Resolution Geometry

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Abstract

The first Assembly paper reconciled the top-down quotient and the bottom-up automorphism groupoid through six independent gates. Each later module—observation laws, ignorance, operational duration, physical distance—added gates of its own. A reader could accept any subset, so Resolution Geometry (RG) risked being a catalogue of rungs rather than one structure. This companion paper removes that freedom. All protocols of a physical slice (source, clock, radar) are one joint observation law F_{tot} ; its kernel groupoid $\mathcal{K}$ is the single identification object; its covariance group Stab_Ψ and the exact sequence $1 \rightarrow \text{Gau} \rightarrow \text{Stab}_\Psi \rightarrow \Gamma \rightarrow 1$ produce one effective symmetry group Γ acting faithfully on the quotient. We prove that every *natural* module (a pullback of a relabeling-invariant structure on the law space) is automatically kernel-basic and Γ -invariant, so basicness, representative invariance and projectability stop being hypotheses. The remaining content of every module is representation theory of one group: clock charts with equal kernels form a groupoid with forced cocycle closure, affine under a time flow; a Lorentz subgroup of Γ forces the quadratic cone; diagonal Euclidean gauge forces a one-dimensional two-point invariant; relative Euclidean covariance forces $g = \lambda^2 h$ with λ constant; a counted channel, having trivial order automorphisms, removes dilations from Γ and fixes the metre. The six Orbit–Cone gates and the six physical-distance gates collapse to three items: the content of Γ , regularity, and statistical validation. A slice therefore stands or falls as a whole. The hostile battery (seed 20260929) passes 15/15 checks, falsifies 14 naive claims with counterexamples, survives seeds 1, 42, 777, 2026, and a canary run shows each of four sabotaged gears is caught.

Contents

1	What was naive in the first assembly	2
2	The joint law and its kernel	2
3	The stabilizer and the exact sequence	3
4	Naturality: why the gears must meet at the kernel	3
5	How the declared groupoid $A(G)$ climbs	4
6	Duration from the kernel	4
7	Cone and scale from Γ	5
8	Physical distance from Γ	5
9	The Kernel–Stabilizer Theorem	6

10 Gate collapse	6
11 Ladder insertion (Stratum 8)	7
12 Reproducibility and self-attack ledger	7
13 Non-claims	8
14 Conclusion	8

1 What was naive in the first assembly

The Observable Orbit–Cone Theorem [1] assumed: (1) gauge soundness, (2) orbit completeness, (3) saturation, (4) transport projectability, (5) cone projectability, (6) a quadratic cone, plus an independent integrable scale. The physical-distance paper [7] added G1–G6. None of these hypotheses was connected to any other. Three defects follow.

1. **Protocol isolation.** Each module was defined on its own protocol and its own quotient. Nothing guaranteed that the Fisher quotient, the clock quotient and the radar quotient were quotients of one latent space by one relation.
2. **Gate inflation.** Projectability and basicness were checked module by module, as if they could fail independently for constructions that are all computed from observed laws.
3. **Symmetry without a home.** Isotropy, homogeneity, no-redshift and “the same group transports the radar” were stated as physical gates, but no object carried them jointly.

The repair is one sentence: *take the joint law of every protocol that the slice uses, and derive every module from its kernel and its stabilizer.*

2 The joint law and its kernel

Definition 2.1 (Physical slice). A physical slice is $\mathcal{S} = (X, \{F_i : X \rightarrow Y_i\}_{i \in I}, \Psi)$: a latent space X , protocols F_i (for example $i \in \{\text{src}, \text{clk}, \text{rad}\}$) valued in spaces of observable laws, and a declared relabeling group $\Psi = \prod_i \Psi_i$, $\Psi_i \leq \text{Aut}(Y_i)$, acting by pushforward along outcome bijections. The joint law is $F_{\text{tot}} = (F_i)_i : X \rightarrow \prod_i Y_i$, with kernel groupoid $\mathcal{K} = X \times_{F_{\text{tot}}} X \rightrightarrows X$ and quotient $\pi : X \rightarrow Q = X/\mathcal{K}$.

Lemma 2.2 (Joint kernel; forced; [3], AK01). $\mathcal{K} = \bigcap_i \mathcal{K}_{F_i}$, and $Q \hookrightarrow \prod_i X/\mathcal{K}_{F_i}$ is injective but generally not surjective. In 300 random finite models the symmetric difference between $\mathcal{K}$ and $\bigcap_i \mathcal{K}_i$ was empty, while the product of quotients exceeded the joint quotient by up to 10 classes (mean 6).

Protocols are therefore coupled through one latent space; their quotients cannot be treated as independent coordinates.

Lemma 2.3 (Garbling enlarges the kernel; forced; AK02). For a parameter-independent Markov kernel M , $\mathcal{K}_F \subseteq \mathcal{K}_{M \circ F}$, strictly in general (merging two outcomes identifies (0.1, 0.3, 0.6) with (0.3, 0.1, 0.6)). The ignorance law “processing cannot reduce ignorance” is a kernel inclusion.

Lemma 2.4 (Count rigidity; forced; AK03). The order automorphism group of a count record $\{0, \dots, N\}$ is trivial (checked exhaustively for $N \leq 7$), and $n \mapsto kn$ is not surjective for $k \geq 2$. Hence $\Psi_i = \{\text{id}\}$ on every counted channel. Continuous readouts carry the positive affine gauge $t \mapsto at + b$.

Lemma 2.4 is the only place where the unit problem touches the root: a counted record cannot be relabeled, so a counted channel cannot be rescaled.

3 The stabilizer and the exact sequence

Definition 3.1 (Covariance, gauge, effective symmetry).

$$\text{Stab}_\Psi(F_{\text{tot}}) = \{(\varphi, \psi) \in \text{Diff}(X) \times \Psi : F_{\text{tot}} \circ \varphi = \psi \circ F_{\text{tot}}\}, \quad \text{Gau} = \{\varphi : F_{\text{tot}} \circ \varphi = F_{\text{tot}}\},$$

with elements of Stab_Ψ identified when their ψ agree on $\text{im } F_{\text{tot}}$. Gauge elements are global bisections of $\mathcal{K}$.

Theorem 3.2 (Kernel exact sequence; forced; AK04). (i) Stab_Ψ normalizes $\mathcal{K}$: $(\varphi \times \varphi)(\mathcal{K}) = \mathcal{K}$. (ii) $\text{Gau} \trianglelefteq \text{Stab}_\Psi$. (iii) $\Gamma := \text{Stab}_\Psi / \text{Gau}$ acts faithfully on Q by $\gamma[x] = [\varphi x]$, giving

$$1 \longrightarrow \text{Gau} \longrightarrow \text{Stab}_\Psi \longrightarrow \Gamma \longrightarrow 1.$$

(iv) The sequence need not split.

Proof. If $F_{\text{tot}}(x) = F_{\text{tot}}(x')$ then $F_{\text{tot}}(\varphi x) = \psi F_{\text{tot}}(x) = \psi F_{\text{tot}}(x') = F_{\text{tot}}(\varphi x')$, which gives (i) and well-definedness of the action. An element acts trivially on Q iff $F_{\text{tot}}(\varphi x) = F_{\text{tot}}(x)$ for all x , iff $\psi|_{\text{im } F_{\text{tot}}} = \text{id}$, iff it lies in Gau ; this gives (ii) and faithfulness. For (iv), let $X = \mathbb{Z}_4$, $F(x) = x \bmod 2$ and declare $G = \mathbb{Z}_4$ acting by rotation. Then $G \cap \text{Gau} = \{0, 2\}$ and the image in $\text{Aut}(Q)$ is $\mathbb{Z}_2$, but the only subgroup of order two is $\{0, 2\}$ itself, so no complement exists. Brute force gives $|\text{Stab}_\Psi| = 8$, $|\text{Gau}| = 4$, $|\Gamma| = 2$ for $\Psi = \text{Sym}(Y)$. $\square$

Remark 3.3. Non-splitting is not a defect: an observable symmetry of Q need not lift to a latent symmetry of the same order. Latent reconstruction is exactly the question whether a splitting exists, which the quotient does not decide.

4 Naturality: why the gears must meet at the kernel

Definition 4.1 (Natural module). A module is *natural* if it is a pullback $M_F = F^*\mu$ of a Ψ -invariant structure μ on the law space: a function (observable invariant, clock reading, radar count), a tensor (Fisher–Rao metric restricted to the family), a cone field ($C_x = DF_x^{-1}(C_{F(x)}^Y)$ for an admissibility cone on laws), or a cocycle of such functions.

Theorem 4.2 (Natural descent; forced; AK05). Let π be a submersion on a regular stratum with $F_{\text{tot}} = \bar{F} \circ \pi$. For every natural module:

1. $M = \pi^*\bar{M}$ with $\bar{M} = \bar{F}^*\mu$ unique; in particular M annihilates $V^{\text{obs}} = \ker D\pi$ and is representative invariant;
2. $\bar{M}$ is Γ -invariant;
3. for tensors, $\text{rad } M \supseteq V^{\text{obs}}$, with equality on strata where $\bar{F}$ is an immersion.

Proof. (1) $F^*\mu = (\bar{F} \circ \pi)^*\mu = \pi^*\bar{F}^*\mu$. (2) For $(\varphi, \psi) \in \text{Stab}_\Psi$, $\varphi^*F^*\mu = (\psi \circ F)^*\mu = F^*\psi^*\mu = F^*\mu$. (3) is (1) plus immersivity. $\square$

Corollary 4.3 (Gates that were never gates). For natural modules, kernel-groupoid basicness [6], transport projectability of law-defined transports and cone projectability of law-defined cones [1] hold automatically. The ignorance law $\text{rad } g_F = V^{\text{obs}}$ [3] is item (3). A measured basicness failure is therefore a diagnosis: the construction used latent structure that the law does not carry.

In the curved-fibre test $q = (a + \frac{1}{2}h^2, b)$ the Fisher metric annihilated the fibre tangent to 2×10^{-15} and agreed across representatives to 1.4×10^{-14} , whereas the latent Euclidean norm of the orthogonal lift, a non-natural construction, differed by 2.73.

Remark 4.4 (What stays a real gate). A transport W generated by latent dynamics rather than by the law is not natural, and its projectability remains a genuine hypothesis. This is where the largest open rung (a canonical connection, R27) lives.

5 How the declared groupoid $A(G)$ climbs

The declared data are a subgroupoid $A \subseteq \mathcal{K}$ (soundness is now membership) and a declared physical group $G \leq \text{Stab}_\Psi$ with image $\Gamma_G \leq \Gamma$.

Proposition 5.1 (Coherence; AK06). *If $gAg^{-1} = A$ for all $g \in G$, the completeness locus $\{x : A \cdot x = [x]\}$ is G -invariant; if Γ_G is transitive on a stratum, completeness at one point is completeness on the whole stratum. Without coherence, completeness is frame dependent.*

Proof. $g(A \cdot x) = (gAg^{-1})(gx) = A \cdot gx$ and $g[x] = [gx]$ by Theorem 3.2(i). $\square$

On $X = \mathbb{Z}_6 \times \mathbb{Z}_2$ with a silent bit, bit flips declared only at even positions are sound, complete at 0, 2, 4, incomplete at 1, 3, 5, and their conjugate under the shift is flips at odd positions. The G -stable choice is complete everywhere.

Rung / module	What A and G do	What is forced
Root (R1–R6)	$A \subseteq \mathcal{K}$; comparison $X/A \rightarrow Q$	$\mathcal{K}$ exact; orbit equality iff completeness
Shadows (R7–R11)	frame gauge $O(p) \times O(q)$ is Ψ on law space	invariants = Ψ -invariant μ , natural
Ignorance	V^{obs}/V^A is the silent defect	$\text{rad } g = V^{\text{obs}}$ on regular strata
Duration	clock charts; time flow in Stab_Ψ	cocycle forced; affine under flow (Thm. 6.1)
Cone	isotropy Γ_q on $T_q Q$	Lorentz cone and $[g]$ if $SO^+(1, n) \leq \Gamma_q$, $n \geq 2$
Scale	count channel in F_{tot}	dilations leave Γ (Lemma 2.4)
Distance	$E(n)$ diagonal in Gau, relative in Γ	one invariant; $g = \lambda^2 h$, λ const.

Table 1: The declared groupoid and group at each rung. Every right-hand entry is a consequence of Theorems 3.2–4.2 plus the content of Γ .

6 Duration from the kernel

Let $T_i = \tau_i \circ F_{\text{clk}, i}$ be the readouts of clock families on the history stratum.

Theorem 6.1 (Forced cocycle; AK07). *(i) If $\mathcal{K}_{T_i} = \mathcal{K}_{T_j}$, there is a unique bijection $f_{ji} : T_i(X) \rightarrow T_j(X)$ with $T_j = f_{ji} \circ T_i$, and the family $\{f_{ji}\}$ is a groupoid: $f_{ii} = \text{id}$, $f_{ij} = f_{ji}^{-1}$, $f_{ki} = f_{kj} \circ f_{ji}$. If $\mathcal{K}_{T_i} \neq \mathcal{K}_{T_j}$, no such map exists. (ii) If a flow $\Phi_s \in \text{Stab}_\Psi$ satisfies $T_i \circ \Phi_s = T_i + a_i s$ with $a_i > 0$ and the $T_i(X)$ are intervals, every f_{ji} is affine with slope a_j/a_i , unique up to the unit of $R29$. (iii) Path-independent universal accumulation H_{PI} of [5] holds iff $\mathcal{K}_{\Delta t} \subseteq \mathcal{K}_T$ on the space of histories.*

Proof. (i) Universal factorization through $X/\mathcal{K}_{T_i} = X/\mathcal{K}_{T_j}$; uniqueness makes composition closed. (ii) $f_{ji}(T_i(\Phi_s x)) = T_j(\Phi_s x)$ gives $f_{ji}(t + a_i s) = f_{ji}(t) + a_j s$ for all t in the interval and all admissible s , i.e. $f_{ji}(t + u) - f_{ji}(t) = (a_j/a_i)u$: f_{ji} is affine with slope a_j/a_i . (iii) H_{PI} asserts that T factors through Δt . $\square$

Numerically: cocycle defect 10^{-15} for three clocks sharing a kernel; the nonlinear clock $\tau + 0.05\tau^2$ breaks translation covariance (0.90) and its transition map misses the best affine fit by 0.81; an environment-dependent clock has non-functional transitions (1.65). The held-out cocycle tests of [5] are therefore tests of kernel equality, and its path-dependence theorem is a failure of a kernel inclusion.

7 Cone and scale from Γ

Theorem 7.1 (Lorentz forcing; AK12). *Let the isotropy Γ_q act on $T_q Q \cong \mathbb{R}^{1+n}$ and contain $SO^+(1, n)$ in its standard representation, $n \geq 2$. Then every Γ_q -invariant closed, convex, pointed cone with nonempty interior is the future or past light cone, and the invariant symmetric bilinear forms are $\{\lambda\eta\}$. Hence the quadratic gate of [1] and the conformal class $[g]$ are forced. For $n = 1$ the spacelike wedge $x \geq |t|$ is an additional invariant convex cone.*

Proof. An invariant cone is a union of orbits: timelike hyperboloid sheets, null half-cones, one-sheeted spacelike hyperboloids and 0. For $n \geq 2$ a one-sheeted hyperboloid contains v and $-v$, which pointedness excludes; convexity and closedness then leave one closed light cone. The standard representation is absolutely irreducible, so invariant forms are one-dimensional. $\square$

Numerically: boosted light-cone boundaries stayed on the cone to 2×10^{-15} ; the quartic cone $t \geq (x^4 + y^4)^{1/4}$ was violated by 1.34; the invariant-form space had dimension 1. Lorentz–Finsler, multicone and characteristic branches (R19–R21) occur exactly when boosts are absent from Γ .

Proposition 7.2 (Anchor is kernel intersection; AK08, AK13). *Imaging in pixel units with unknown pitch has Fisher rank 3 in (p_x, p_y, s, σ) with the dilation as exact null direction; adding the radar count to F_{tot} gives rank 4. A physical dilation is in Stab_Ψ for the imaging channel alone (defect 8×10^{-14} with outcome rescaling) and outside Stab_Ψ once the counted channel is present (defect 164 noise units). The unit is fixed exactly when dilations leave Γ .*

Proposition 7.3 (No-redshift is Γ -membership; AK09). *With a clock-rate gradient κ in the clock channel, rotations belong to Stab_Ψ iff $\kappa = 0$ (defect 2×10^{-13} versus 228 noise units at $\kappa = 10^{-3}$). The gate G_4 of [7] is not an extra physical assumption: it is the statement that the relevant rotations and translations are in the stabilizer of the joint law.*

8 Physical distance from Γ

Theorem 8.1 (Two-point forcing; AK10). *Suppose X contains station and target configurations in $\mathbb{R}^n$, $n \geq 2$, and the diagonal action of $E(n)$ lies in Gau . Then the orbit space of pairs is $[0, \infty)$, and every natural two-point module is a function of the Euclidean distance ρ . If the radar count is additive along concatenations that are collinear in the sense of equality in the triangle inequality for ρ , $\rho = (c/2\Delta\nu_{\text{Cs}})n_{\text{Cs}}$.*

Proof. $E(n)$ is transitive on pairs at fixed distance (explicitly: rotate the difference vector, then translate); an invariant function is constant on orbits. Additivity plus monotonicity gives linearity (Cauchy); the constant is fixed by the counted channel and the SI definition. $\square$

With translations alone the orbit space is two-dimensional (a displacement vector survives) and distance is not forced; for $E(2)$ the computed orbit-space dimension is 1 and random pairs at equal distance were connected with error 2×10^{-15} .

Theorem 8.2 (Metric forcing; AK11, AK14). *If in addition relative Euclidean motions of the target lie in Γ with irreducible isotropy, then for every natural metric module g on the target chart, $g = \lambda^2 h$ where h is the unique invariant metric with distance ρ , and λ is constant.*

Proof. g and h are Γ -invariant (Theorem 4.2); Schur gives proportionality at each point; the ratio is a Γ -invariant function, constant by transitivity. $\square$

Measured: λ spread 1.9×10^{-10} in a homogeneous slice, 0.49 with $\sigma(p) = \sigma_0(1 + 0.2p_x)$. The isometry algebras of the imaging Fisher metric and the radar metric coincide for an isotropic detector and differ by 0.41 for axis ratio 1.6: the mismatch is detected from data, without external truth.

9 The Kernel–Stabilizer Theorem

Theorem 9.1 (Kernel–Stabilizer Theorem). *Let $\mathcal{S}$ be a physical slice with joint law F_{tot} , kernel groupoid $\mathcal{K}$, and effective symmetry $\Gamma = \text{Stab}_\Psi/\text{Gau}$. Assume every module is natural and work on a regular stratum. Then:*

1. *every module descends to Q and is Γ -invariant; exact ignorance is $\mathcal{K}$ and infinitesimal ignorance is $V^{\text{obs}} = \text{rad } g$;*
2. *a declared $A \subseteq \mathcal{K}$ is sound; if G -coherent, completeness is a property of Γ_G -orbits;*
3. *clocks with equal kernels form one duration object with forced cocycle, affine under a time flow in Stab_Ψ ; H_{PI} is a kernel inclusion;*
4. *if $SO^+(1, n) \leq \Gamma_q$, $n \geq 2$, the cone is Lorentzian and $[g]$ is forced;*
5. *if a counted channel is in F_{tot} , dilations are not in Γ ; if moreover diagonal $E(n) \leq \text{Gau}$ and relative $E(n) \leq \Gamma$ with irreducible isotropy, then $g_{\mathcal{E}} = \lambda_{\mathcal{E}}^2 h$ with $\lambda_{\mathcal{E}}$ constant and h the unique metric whose distance is the radar distance in metres.*

Corollary 9.2 (Joint falsifiability). *All module predictions are consequences of one empirical object, Γ_{emp} , estimated by equality-of-law tests alone. Any failed prediction implies one of: (a) Γ is smaller than declared, (b) the data leave a regular stratum, or (c) a module is not natural. The cross-module lock equations*

$$\begin{aligned} \text{L1 } \text{rad } g &= V^{\text{obs}}, & \text{L2 } \text{isom}(g) &= \text{isom}(h), & \text{L3 } \text{clock cocycle } &\Gamma\text{-invariant,} \\ \text{L4 } \lambda &\text{ constant,} & \text{L5 } \Gamma_q C_q &= C_q, & \text{L6 held-out anchor} \end{aligned}$$

share Γ , so a single broken gear breaks the others' predictions with it.

In the Monte-Carlo slice (30 points, 4000 draws each, Bonferroni $z_{\text{crit}} = 4.46$) the homogeneous slice retained rotations ($\max |z| = 3.09$), while a redshift (891), a 2% detector anisotropy (1786) and a 5% noise gradient (23.8) each removed the same generator. Seeds 1, 42, 777, 2026 gave $\max |z|$ between 2.28 and 3.63 for the homogeneous slice.

10 Gate collapse

Former gate	Status under Theorem 9.1
Orbit–Cone (1) soundness	definition: $A \subseteq \mathcal{K}$
Orbit–Cone (2) completeness	Γ_G -orbit property for coherent A (Prop. 5.1)
Orbit–Cone (3) saturation	still a gate: coordinates on Q , discrete residuals possible
Orbit–Cone (4) transport projectability	automatic for law-defined transport; gate for latent W
Orbit–Cone (5) cone projectability	automatic for law-defined cones
Orbit–Cone (6) quadratic cone	forced by $SO^+(1, n) \leq \Gamma_q$, $n \geq 2$
Integrable scale	count channel + transitive Γ
PD G1 covariance, irreducible isotropy	content of Γ
PD G2 regularity	regular stratum (kept)
PD G3 radar transported by the same group	automatic: Stab_Ψ is the stabilizer of the joint law
PD G4 no redshift	content of Γ on the clock channel
PD G5 medium speed anchored	kernel intersection: unanchored v puts (d, v) back in $\mathcal{K}$
PD G6 held-out anchor	statistical validation (kept)

Table 2: Thirteen hypotheses reduce to: the content of Γ , regularity, saturation, naturality of the modules, and statistical validation.

11 Ladder insertion (Stratum 8)

A new type is added: Γ -forced means forced once the stated subgroup of Γ is established by equality-of-law tests.

Rung	Statement	Type	Depends / attack
R45	Joint law of all protocols; $\mathcal{K} = \bigcap \mathcal{K}_i$	forced	R1, AK01
R46	Garbling enlarges the kernel	forced	R14, AK02
R47	Count rigidity: Ψ trivial on counted channels	forced	AK03
R48	Exact sequence $1 \rightarrow \text{Gau} \rightarrow \text{Stab}_\Psi \rightarrow \Gamma \rightarrow 1$; Γ faithful on Q ; may not split	forced	R4, AK04
R49	Natural modules are basic and Γ -invariant; $\text{rad } g = V^{\text{obs}}$	forced	R8, R11, AK05
R50	Coherent A : completeness is a Γ_G -orbit property	gated (coherence)	R6, AK06
R51	Equal clock kernels $\Rightarrow$ forced cocycle; time flow $\Rightarrow$ affine; $H_{\text{PI}} = \text{kernel inclusion}$	forced / Γ -forced	R28–R29, AK07
R52	Anchor = kernel intersection; dilations leave Γ iff a counted channel is present	forced	R38–R39, AK08, AK13
R53	No-redshift = rotations/translations in Stab_Ψ of the joint law	Γ -forced	R40, AK09
R54	Lorentz subgroup forces the light cone and $[g]$ ($n \geq 2$)	Γ -forced	R17–R18, AK12
R55	Diagonal $E(n)$ gauge forces one two-point invariant	Γ -forced	AK10
R56	Relative $E(n)$ with irreducible isotropy forces $g = \lambda^2 h$, λ const.	Γ -forced	R36, AK11, AK14
R57	Kernel–Stabilizer Theorem and lock equations L1–L6	Γ -forced	R45–R56, AK15
R58	Canonical connection for non-natural latent transport	open	R27

12 Reproducibility and self-attack ledger

Run	Result
seed 20260929	15/15 pass; 14 naive claims falsified with explicit counterexamples
seeds 1, 42, 777, 2026	15/15 each
--canary	AK07, AK09, AK11, AK12 sabotaged; only the targeted check fails in each run
self-attack 1	AK13 was first registered with an unjustified threshold (10^3 noise units) and failed at 164. The counterexample was unchanged; the gate was re-registered at 10 noise units, the standard used for all other defects. Recorded, not hidden.
self-attack 2	the first AK12 canary was <i>not</i> caught: the cone boundary sampler was decoupled from the cone definition. Cones are now generated from one spatial norm; the canary is caught.

13 Non-claims

Non-claim 13.1. The content of Γ is empirical. The theorem says what each subgroup forces, not that a given laboratory possesses it.

Non-claim 13.2. The relabeling group Ψ on continuous channels is declared. Count rigidity constrains counted channels only.

Non-claim 13.3. Non-natural modules, in particular transports generated by latent dynamics, remain gated; the canonical connection (R27, R58) is open.

Non-claim 13.4. Curved and non-static spacetimes, where rotations or translations leave Γ , and quantum protocols are outside the tested slice. No field equation is derived.

14 Conclusion

The first Assembly paper showed that the quotient and the groupoid meet at the kernel. This paper shows that everything else meets there too, and for one reason: every module RG uses is computed from observed laws, so it is a pullback along the joint law, hence kernel-basic and invariant under the one group that stabilizes that law. Ignorance is the kernel. Observable laws are its invariants. Duration is the groupoid of clock charts with equal kernels. The cone is the orbit geometry of its Lorentz part. Physical distance is the two-point invariant of its Euclidean part, fixed in metres by the one channel that cannot be relabeled. What is left to nature is the content of Γ . That is also what makes the structure falsifiable as a single object: remove one generator and every module that depends on it falls at once.

References

- [1] M. Rada, *Resolution Geometry as an Observable Orbit–Cone Structure*, 2026.
- [2] M. Rada, *The Theorem Ladder of Resolution Geometry*, 2026.
- [3] M. Rada, *The Law and Boundary of Observational Ignorance*, 2026.
- [4] M. Rada, *Observation Laws, Universal Quotients, and Saturated Observable Geometry*, 2026.
- [5] M. Rada, *Operational Duration*, 2026.
- [6] M. Rada, *Distance and Angle Laws of Resolution Geometry*, 2026.
- [7] M. Rada, *From Statistical Experiment to the Metre*, 2026.
- [8] M. J. Pflaum, H. Posthuma, X. Tang, “Geometry of orbit spaces of proper Lie groupoids,” *J. reine angew. Math.* 694 (2014).
- [9] N. N. Čencov, *Statistical Decision Rules and Optimal Inference*, AMS, 1982.
- [10] K. C. H. Mackenzie, *General Theory of Lie Groupoids and Lie Algebroids*, CUP, 2005.
- [11] D. Malament, *J. Math. Phys.* 18 (1977); S. Hawking, A. King, P. McCarthy, *J. Math. Phys.* 17 (1976).
- [12] BIPM, *The International System of Units*, 9th ed., 2019.